

AI Learning Roadmap

From Beginner to Expert

A structured 7-phase path covering foundations, deep learning, transformers, specialisation, and expert practice.

Phase 1 — Foundations

Build the mathematical and programming base essential for all AI work. Estimated time: 2–4 months.

Mathematics Linear algebra, calculus, probability	Python NumPy, Pandas, Matplotlib	Statistics Probability, inference, distributions	Data skills SQL, EDA, data cleaning
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Recommended resources:

- 3Blue1Brown — Essence of Linear Algebra (YouTube)
- Khan Academy — Multivariable Calculus
- Python for Data Analysis — Wes McKinney (book)
- fast.ai Practical Data Ethics course

Phase 2 — Classical Machine Learning

Master the foundational algorithms that power most production ML systems. Estimated time: 2–3 months.

Supervised learning Regression, classification, SVMs	Unsupervised Clustering, PCA, k-means	Model evaluation Cross-validation, metrics, bias-variance	Scikit-learn Pipelines, hyperparameter tuning
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Recommended resources:

- Hands-On Machine Learning — Aurélien Géron (book)
- fast.ai — Practical Machine Learning course
- Kaggle — Learn tracks (free, interactive)
- Andrew Ng — Machine Learning Specialization (Coursera)

Phase 3 — Deep Learning

Learn neural networks and modern deep learning frameworks used in industry. Estimated time: 3–4 months.

Neural networks Backprop, activations, optimizers	CNNs ResNet, EfficientNet, vision tasks	RNNs / LSTMs Sequences, time-series, NLP basics	PyTorch / TF Training loops, GPUs, checkpointing
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Recommended resources:

- Deep Learning — Goodfellow, Bengio, Courville (book)
- fast.ai — Practical Deep Learning for Coders
- CS231n — Convolutional Neural Networks (Stanford)
- PyTorch official tutorials and documentation

Phase 4 — Transformers & Modern AI

Master the architecture behind GPT, DALL-E, Claude, and most state-of-the-art models. Estimated time: 2–3 months.

LLMs / GPT Self-attention, RLHF, prompting	Diffusion models Image generation, CLIP, DALL-E	Multimodal AI Vision-language, embeddings	Fine-tuning LoRA, QLoRA, PEFT, adapters
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Recommended resources:

- Attention Is All You Need — Vaswani et al. (original paper)
- Hugging Face — NLP Course (free)
- Andrej Karpathy — Let's build GPT from scratch (YouTube)
- d2l.ai — Dive into Deep Learning (free textbook)

Phase 5 — Pick a Specialisation

Go deep in one domain to build expert-level skills and stand out. Estimated time: 3–6 months.

NLP LLMs, RAG, AI agents	Computer vision Detection, segmentation, 3D	Reinforcement learning PPO, DQN, RLHF, sim	MLOps Deploy, monitor, CI/CD, infra
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Recommended resources:

- Spinning Up in Deep RL — OpenAI (RL track)
- Full Stack Deep Learning — deployment focus
- LangChain / LlamaIndex docs — agents and RAG
- AWS/GCP/Azure ML certifications — MLOps track

Phase 6 — Advanced & Research

Engage with cutting-edge research and develop original contributions to the field. Estimated time: ongoing.

AI safety Alignment, interpretability, RLHF	AI agents Planning, tool use, multi-agent	Foundation models Scaling laws, emergent abilities	Research skills arXiv, replication, publishing
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Recommended resources:

- Papers With Code — track state of the art
- arXiv cs.LG / cs.AI — daily papers
- Anthropic, DeepMind, OpenAI research blogs
- Replicate 1–2 papers per month to build depth

Phase 7 — Expert Practice

Build your reputation, portfolio, and career path as a practising AI expert. Estimated time: ongoing.

Projects Portfolio, Kaggle, GitHub	Open source HuggingFace, PyTorch, LangChain	Publish NeurIPS, ICML, ICLR, arXiv	Career paths Researcher, engineer, product
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Recommended resources:

- Kaggle competitions — build a public track record
- Contribute to HuggingFace Transformers or similar OSS
- Write blog posts explaining your projects (Medium, Substack)
- Attend NeurIPS, ICML, ICLR — network and present

Key Platforms & Communities

fast.ai Free practical deep learning courses	deeplearning.ai Andrew Ng specializations	Hugging Face Models, datasets, spaces	Kaggle Competitions and datasets
d2l.ai	arXiv	Papers With Code	CS231n / CS224n

AI Learning Roadmap

Interactive deep learning textbook	Preprints — cs.LG and cs.AI	SOTA benchmarks & code	Stanford vision & NLP courses
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Total estimated time: 18–36 months for full roadmap. Pace varies by background and dedication.